MIL-R-35084A

16 June 1966

SUPERSEDING
MIL-R-35084
29 June 1962

MILITARY SPECIFICATION

RICE, INSTANT, ENRICHED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers enriched, instant rice for use by the Armed Forces as an item of general issue and as a component of operational rations. The product is a modified whole grain rice that can be prepared for use by the addition of boiling (or near boiling) water without additional cooking.

1.2 Classification.- The product shall be of the following types, as specified (see 6.1):

Type I - Instant rice
Type II - Low moisture instant rice

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Box, Fiberboard
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter

MILITARY

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaperproof or Waterproof, Flexible
MIL-L-35078	-	Loads, Unit; Preparation of Nonperishable Subsistence In

/FSC 8920/

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-668 - Sanitary Standards for Food Plants

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply:

U. S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

National Research Council

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, Washington, D. C. 20418.)

United States Pharmacopoeial Convention, Inc.

Pharmacopoeia of the United States

(Application for copies should be addressed to the United States Pharmacopoeial Convention, Inc., 46 Park Ave., New York, N. Y. 10016.)

Association of Official Agricultural Chemists

Official Methods of Analysis

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D. C. 20044.)

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Association of Vitamin Chemists

Methods of Vitamin Assay. (2nd Edition)

(Application for copies should be addressed to Interscience Publishers, John Wiley & Sons, Inc., 605 Third Ave., New York, New York 10016.)

3. REQUIREMENTS

3.1 Material.- All materials shall be sound, wholesome, and free from evidence of insect infestation or other objectionable foreign matter.

3.1.1 Rice.- Rice shall be whole grain milled rice.

3.1.2 Enrichment ingredients.- Enrichment ingredients shall be thiamine, niacin or niacinamide, iron and riboflavin, of a grade conforming to the Pharmacopoeia of the United States, or Food Chemicals Codex.

3.2 Finished product.-

3.2.1 Physical requirements.- The finished product (type I or II) shall be white in color. It shall be free from rancid, musty, bitter or any other undesirable flavors or odors. The product shall meet performance test specified in 4.5.2.

3.2.2 Analytical requirements.- The finished product (type I and II) shall have a protein content of not less than 5.0 percent, and ash content of not more than 0.70 percent, both calculated on a moisture-free basis. The moisture content for type I product shall be not more than 12.0 percent, and for type II product, not more than 6.0 percent.

3.2.3 Enrichment requirements.- The product shall comply with the requirements listed in table I, based on lot average requirements.

TABLE I.- Enrichment ingredients

Ingredients	Content per pound	
	Minimum	Maximum
	Milligrams	
Thiamine	2.0	4.0
Niacin or niacinamide	16.0	32.0
Iron	13.0	26.0
Riboflavin	1.2	2.4

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3.3 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.4 Workmanship.- The product shall be processed in establishments meeting the requirements of MIL-STD-668. It shall be processed with minimum delay between the various stages in production. The sealed and finished packages shall be clean and free from defects such as, springers, flippers, swells, dents, rust and improper closures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection (plant sanitation inspection).- Award of contracts shall be limited to plants which have received prior sanitary approval by the Military Veterinary Services in accordance with MIL-STD-668 or to plants operating under the continuous inspection supervision of the Processed Products and Standardization Branch, Consumer and Marketing Service, U. S. Department of Agriculture and as officially listed by that agency.

4.3 Sampling for inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated herein.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of rice.- Examination of rice shall be performed organoleptically, as necessary, to determine conformance to 3.1 and 3.1.1. The sample for examination shall be a 5-pound composite of product representative of the lot. When bags of product are involved, the composite shall be derived from 10 Bags of rice. Any nonconformance shall indicate an unacceptable component and use of such shall be cause for rejection of the involved quantity of finished product.

4.3.1.2 Enrichment ingredients.- Conformance to 3.1.2 shall be ascertained by examination of pertinent labels, invoices or other valid documents.

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Use of nonconforming enrichment ingredients shall be cause for rejection of the involved finished product.

4.3.1.3 Testing of cans.- Cans shall be tested in accordance with the criteria in table II. Requirements for tin plate shall be a lot average requirement. Nonconformance to tin plate requirements shall indicate unacceptable cans.

TABLE II.- Testing of cans

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test reference and requirement paragraph
Cans 1/	1 can w/lid	Cans	S-1	Tin plate	Nearest 0.01 lb/ base box	5.1.1.2 and 4.5.3

1/ Test for can coating (when required) and label shall be performed on the same samples submitted for tin plate test. One or more test failures shall indicate unacceptable cans.

4.3.2 Examination of end item.- The end item shall be examined in accordance with tables IV through VII and 4.3.2.1 through 4.3.2.5. Sampling plans and acceptance criteria shall be as shown in table III. The acceptable quality levels (AQLs) shall be expressed in terms of defects per hundred units.

TABLE III.- Sampling plan and acceptance criteria

Tables	Inspection level	Sample unit	Lot size expressed in	AQLs		
				Major	Minor	Total
IV	I	1 carton	Cartons	1.5	---	4.0
V	S-3	1 primary container	Primary containers	---	2.5	---
VI	S-3	1 intermediate fiberboard container	Intermediate fiberboard containers	---	6.5	---
VII	S-2	1 primary container	Primary containers	1.5	---	---

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TABLE IV.- Examination of carton

Category		Defect
Major	Minor	
151		Hole or tear, or not securely closed (sifter).
	201	Not clean.
	202	Objectionable odor.
152		Nomenclature missing, incorrect or illegible.
	203	Required label information, other than nomenclature, missing, incorrect or illegible.

TABLE V.- Examination for net weight 1/ 2/

Category	Defect
Minor	
201	Net weight more than 5 percent under specified net weight (applicable to containers required to have 1 pound or less).
202	Net weight more than 2 percent under specified net weight (applicable to containers required to have more than 1 pound).

1/ Report results to nearest 1/8 ounce for containers required to have 1 pound or less, and nearest 1/4 ounce for containers required to have more than 1 pound.

2/ Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight.

TABLE VI.- Examination of intermediate fiberboard container

Category	Defect
Minor	
201	Not RSC style.
202	Not securely closed by tape or glue.
203	Manufacturer's joint not taped.

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TABLE VII.- Examination of product 1/

Category	Defect
Major	
	Dehydrated
151	Not white.
	Rehydrated (sec 4.5.2) 2/
152	Not completely rehydrated (i.e., presence of hard grains).
153	Undesirable flavor or odor.

1/ Presence of foreign material (e.g., insects, insect parts, filth, glass, metal, paint, stones, etc.) shall be cause for rejection of the lot.

2/ Make one determination per sample unit.

4.3.2.1 Examination of case liners or overwraps.- Examination of case liners or overwraps shall be in accordance with the examination criteria contained in the appendix to MIL-L-10547.

4.3.2.2 Examination of cans.- Examination of cans for exterior condition and seam leakage shall be in accordance with the applicable examination criteria contained in PPP-C-29.

4.3.2.3 Examination of can coating.- When exterior can coating is required to be in accordance with TT-C-495, examination of can coating shall be in accordance with the examination criteria contained in that specification.

4.3.2.4 Examination of can labeling.- Examination of can label shall be in accordance with the examination criteria of MIL-L-1497.

4.3.2.5 Examination of shipping containers.- Shipping containers shall be examined in accordance with the examination criteria contained in the appendix to PPP-B-636. In addition, the following defects shall be included in the examination: Major - Markings missing, incorrect or illegible; Minor - Arrangement of cans not as specified; Minor - Marking indicates total net weight of product, in cartons, exceeds 40 pounds. For level C pack, only the defect applicable to marking shall apply.

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4.3.3 Sampling plans and acceptance criteria for testing finished product.- The finished product shall be tested for the analytical and enrichment requirements of 3.2.2 and 3.2.3. Procedure shall be in accordance with 4.5. Test requirements shall be lot average requirements. The sample for testing shall be a 1-pound composite of product derived from the number of containers indicated by inspection level S-2. Lot size shall be expressed in terms of containers. Results shall be reported to the same decimal place or unit as specified for the specific requirement. One or more test failure(s) shall be cause for rejection of the lot.

4.4 Standby test samples.- The government reserves the right to withdraw and hold standby test samples of components or finished product or both (quantity of which shall be not more than twice that required by this specification) for inspection purposes. Samples not used will be returned to the supplier.

4.5 Tests.- Test procedures and controls which differ from those specified herein may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.5.1 Chemical analyses.-

4.5.1.1 Moisture, ash and protein.- Chemical analyses for moisture, ash and protein shall be made in accordance with Official Methods of Analysis of the Association of Official Agricultural Chemists, chapter: Cereal Foods: section: Wheat, Rye, Oats, Corn, Buckwheat, Rice and Barley and Their Products except Cereal Adjuncts. Chemical analysis for iron shall be made in accordance with aforementioned publication, chapter: Cereal Foods, section: Wheat Flour.

4.5.1.2 Thiamine and niacine or niacinamide and riboflavin.- Determination of thiamine and niacin or niacinamide and riboflavin shall be made in accordance with the following methods described in Methods of Vitamin Assay of the Association of Vitamin Chemists; 2nd edition:

Test	Method
Thiamine	Thiochrome (p. 111-123)
Niacin or niacinamide	Microbiological (p. 180-184)
Riboflavin	Microbiological (p. 159-166)

4.5.2 Performance test.- The product shall be completely rehydrated and ready to eat when prepared as follows: to 3 ounces, by weight, of instant rice, add 10 fluid ounces of boiling (or near boiling point) water

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containing 3/4 teaspoon of salt. Stir lightly with a fork, cover, and let stand for 10 minutes. The prepared product shall be hydrated, soft, and shall have an acceptable flavor for immediate eating.

4.5.3 Tin coating weights.- Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- The product shall be packaged in accordance with 5.1.1 or 5.1.2, as specified (see 6.1).

5.1.1 Level A.-

5.1.1.1 Cartons (type I product only).- Twenty-five or 40 ounces of the product, as specified (see 6.1), shall be packaged in a commercial carton of the type normally used for the product. When specified (see 6.1), the product shall be packaged in commercial cartons in other net weights as specified in the contract or order.

5.1.1.1.1 Intermediate container.- Not more than 40 pounds of the product, packaged as specified in 5.1.1.1, shall be packed in a non-test, regular slotted-style corrugated fiberboard container, with a taped manufacturer's joint. Closure shall be effected by taping or gluing.

5.1.1.1.2 Waterproof case liner or overwrap.- The intermediate container shall be enclosed in a waterproof case liner or overwrap, fabricated and sealed in accordance with type I, II or III, grade C, for subsistence items of MIL-B-131547.

5.1.1.2 Cans (type II product only).- Twelve and one-half or 46 ounces of the product, as specified (see 6.1), shall be filled into a size 401 by 411 or 603 by 700, respectively, open-top style, round, metal can, with soldered side seam and compound-lined, double-seamed ends. The can shall be made throughout from commercial electrolytic tin plate of not less than 0.25-pound per base box (see 4.5.3), and shall be coated outside with a coating conforming to type I or, when specified (see 6.1), type III of TT-C-495. The can shall be hermetically sealed and shall not show leakage when tested in accordance with 4.3.2.2.

5.1.2 Level C.- Type I product shall be packaged in accordance with 5.1.1.1. Type II product shall be packaged in accordance with 5.1.1.2, except that cans with or without an exterior coating will be acceptable.

5.2 Packing.- The product shall be packed in accordance with 5.2.1, 5.2.2 or 5.2.3, as specified (see 6.1).

5.2.1 Level A.-

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5.2.1.1 Cartons.- The intermediate container specified in 5.1.1.1.1, and enclosed in the waterproof case liner specified in 5.1.1.1.2, shall be packed in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC-SL, V2s of PPP-B-636.

5.2.1.2 Cans.- Twenty-four size 401 by 411 cans of the product, arranged 4 in length, 3 in width, and 2 in depth, or six size 603 by 700 cans of the product, arranged 3 in length, 2 in width, and 1 in depth, shall be packed on end in a snug-fitting fiberboard shipping container, with sleeve, constructed, closed and reinforced in accordance with style RSC-SL, V2s of PPP-B-636.

5.2.1.3 When specified (see 6.1 and 6.2), the shipping container shall be constructed and closed in accordance with style RSC, V2s of PPP-B-636. The fiberboard boxes shall be palletized and sheathed in accordance with type I, class C of MIL-L-35078. Sheathing and cap shall be of V2s fiberboard.

5.2.2 Level B.- Not more than 40 pounds of the type I product, packaged as specified in 5.1.1.1, or the cans of the type II product, in the quantity and arrangement specified in 5.2.1.2, shall be packed on end in a snug-fitting fiberboard shipping case constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure, of PPP-B-636.

5.2.2.1 When specified (see 6.1 and 6.3), the type I product in an intermediate container specified in 5.1.1.1.1 and enclosed in the waterproof case liner specified in 5.1.1.1.2, or the cans of the type II product, in the quantity and arrangement specified in 5.2.1.2, shall be packed on end in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC, V3c or V3s of PPP-B-636.

5.2.3 Level C.- The product shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carrier applicable to the mode of transportation.

5.3 Labeling and marking.-

5.3.1 Cartons.- Commercial labeling, including directions for preparation supplied by the supplier, that complies with the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder, will be acceptable.

5.3.2 Cans.- Cans shall be labeled in accordance with MIL-L-1497. In addition, the cans shall include the following directions for preparation, as applicable.

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For 401 by 411 cans:

Mix contents of can (12-1/2 oz.) with 1-1/4 qts. of boiling water containing 1-1/2 teaspoons of salt. Stir, cover, and let stand 10 minutes.

Yield: 8 cups of rice.

For 603 by 700 cans:

Mix contents of can (46 oz.) with 4-1/2 qts. of boiling water containing 2 tablespoons of salt. Stir, cover and let stand 10 minutes.

Yield: 28 cups of rice.

5.3.3 Shipping containers.- Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of product required (see 1.2).
(c) Levels of packaging and packing required (see 5.1 and 5.2).
(d) Size of package (carton) required for type I product (see 5.1.1.1).
(e) When product is to be packaged in commercial cartons and in net weight other than specified (see 5.1.1.1).
(f) Can size required for type II product (see 5.1.1.2).
(g) When type III exterior can coating is required (see 5.1.1.2).
(h) When packing specified in 5.2.1.3 or 5.2.2.1 is required.

6.2 Packing specified in 5.2.1.3 is intended to provide the maximum Military protection required for open storage of unit loads in areas of heavy rainfall, high humidity, and adverse storage conditions.

6.3 Packaging specified in 5.2.2.1 is intended to provide the limited Military protection required for shipment of supplies to certain specified overseas destinations, transfer at sea, and for shipment and storage of Special Program Reserves.

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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MIL-R-35084A

Custodians:

Army - GL

Navy - SA

Air Force - 67

Preparing activity:

Army - GL

Project No. 8920-0118

Review activities:

Army - MD

Navy - MC, SA

Air Force - 67

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Documents.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Rice, Instant, Enriched - MIL-R-35084A

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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